

Secrets Hardening Implementation Report

Date: 2026-05-18

Branch: `feature/ai-daily-report`

Summary

Implemented secrets hardening for AI, weather, and eBird settings by adding `apiKeyFile` support, extending secret resolution to return source metadata, updating runtime consumers to resolve from file-or-value safely, and fixing AI settings update normalization behavior.

Commits

- `78e13820` harden secrets: support `apiKeyFile` sources and normalize AI settings updates
- `34c062c6` docs: convert secrets hardening plan into checklist

Files Changed

- `internal/conf/config.go`
- `internal/conf/defaults.go`
- `internal/conf/validate_ai.go`
- `internal/conf/validate_realtime.go`
- `internal/secrets/secrets.go`
- `internal/api/v2/ai.go`
- `internal/api/v2/ai_test.go`
- `internal/api/v2/integrations.go`
- `internal/weather/provider_openweather.go`
- `internal/weather/provider_wunderground.go`
- `internal/conf/config_yaml_tags_test.go`
- `internal/conf/validate_harden_test.go`
- `plan/secrets-hardening-plan.md`

Detailed Changes

1) Config schema: `apiKeyFile` fields

Added `apiKeyFile` to settings structs so users can source secrets from files instead of plaintext config:

- `Realtime.EBird.APIKeyFile`
- `Realtime.Weather.OpenWeather.APIKeyFile`
- `Realtime.Weather.Wunderground.APIKeyFile`
- `AI.APIKeyFile`

Also updated default values to include empty `apikeyfile` keys where applicable.

2) Validation hardening

- AI validation now requires at least one source when enabled:
 - `ai.apiKey` or `ai.apiKeyFile`
- Realtime validation updated similarly:
 - eBird requires `apiKey` or `apiKeyFile` when enabled
 - OpenWeather provider requires `apiKey` or `apiKeyFile`
- Wunderground validation accepts `apiKey` or `apiKeyFile`.

3) Secrets resolver extension

Extended secrets utility with:

- `ResolveWithSource(filePath, value) (secret, source, err)`
- `SecretSource` indicator values (none/file/env-or-text)
- `IsEnvReference(value)` helper for `${...}` detection

This allows callers to both resolve secret values and understand where they came from.

4) Runtime integration updates

Updated runtime consumers to use `ResolveWithSource`:

- AI model listing route (`/api/v2/ai/models`)
- OpenWeather provider
- Wunderground provider
- eBird client initialization path
- weather integration auth test path in API v2 integrations

Added warning logs for plaintext secret usage to support migration to env vars/secret files.

5) AI settings update regression fix

In `UpdateAISettings` (`internal/api/v2/ai.go`):

- Apply normalized output from `ValidateAISettings` before saving.
- Ensures invalid cache hours are normalized (e.g. `0` -> `4`) and persisted correctly.

6) Tests updated

- Added/updated YAML round-trip coverage for new `apiKeyFile` fields.
- Updated validation tests for weather/eBird requirements.
- Added AI route tests and updated expectation text for new key-source validation wording.

Verification Performed

User-ran and confirmed passing:

- `go test github.com/tphakala/birdnet-go/internal/api/v2 -run TestUpdateAISettings -count=1 -v`
- `go test github.com/tphakala/birdnet-go/internal/api/v2 -run "TestGetAISettings|TestUpdateAISettings" -count=1 -v`

Both passed after applying the normalization + assertion fixes.

Notes

- Repository has additional unrelated in-progress changes in working tree; this work was committed in scoped commits.
- Windows line-ending warnings (`LF -> CRLF`) were observed during staging; no functional impact to logic.